

Performance Evaluation of Ultra-Wideband Vivaldi Antenna Arrays with Emphasis on Grating Lobes, and Mutual Coupling

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Abstract— Ultra-wideband (UWB) antennas find extensive use in various wireless communication and radar systems. They offer wide bandwidth and constant radiation characteristics. In this paper, the performance evaluation of Vivaldi antenna arrays at 6 GHz is carried out using MATLAB Antenna Toolbox. Various antenna parameters, including VSWR, radiation pattern, input impedance, mutual coupling, and grating lobes, have been investigated for various antenna array sizes and inter-element spacings. It has been found that when the spacing is 0.8λ , grating lobes do not occur, and VSWR is below 2. However, if the spacing is increased, grating lobes occur. It has also been found that by increasing the number of antenna elements, the antenna gain and directivity can be increased, but at the same time, mutual coupling is increased. It has been found that Vivaldi antenna arrays offer optimal performance.

Keywords—ultra wide band (uwb), Vivaldi antenna, antenna array, grating lobes, mutual coupling, vswr.

I. INTRODUCTION

In 1979, Gibson proposed a Vivaldi antenna as a tapered slot antenna that employs travelling wave radiation to obtain highly directive radiation patterns [1]. The advantages of using a Vivaldi antenna include ultra-wideband, stable radiation performance, and high directivity within its operating frequency band [7]. The Vivaldi antenna is used for various purposes, such as microwave imaging and satellite communication. The advantages of using a Vivaldi antenna cannot be efficiently extended to array configurations. Grating lobes occur when the inter-element spacing is greater than one wavelength, as described by array factor theory and pattern multiplication. For efficient suppression of grating lobes, antenna elements should be closely spaced. This causes increased mutual coupling at lower frequencies, thereby affecting impedance matching and radiation efficiency[5], [9]. Various UWB antennas, such as spiral, sinuous, coupled, and Vivaldi, have been discussed in the literature [2], [4]. The spiral and sinuous UWB antennas, although frequency-independent, have wide beamwidths, high mutual coupling, and poor control of grating lobes for array configurations.

Table 1 shows a comparative analysis of Vivaldi, Spiral, Sinuous, and Coupled Antennas on the basis of various parameters such as VSWR, Return Loss, Radiation

Pattern, Mutual Coupling, Grating Lobes, and Impedance [2], [4], [6], [7], [13].

TABLE 1

Parameter	Vivaldi Antenna	Spiral Antenna	Sinuous Antenna	Coupled Arrays
Structure	Tapered slot design	Spiral geometry	Sinuous arms	Coupled array structure
Bandwidth / S_{11}	$S_{11} < -10$ dB (can reach up to -40 dB)	$S_{11} \approx -10$ to -20 dB	$S_{11} \approx -10$ to -15 dB	$S_{11} \approx -5$ to -15 dB
Gain	4–8 dBi (single), 10–25 dBi (array)	5–8 dBi	3–6 dBi	8–20 dBi
Beamwidth	$>60^\circ$ (single), $<30^\circ$ (array)	70° – 120°	60° – 100°	$<30^\circ$
Mutual Coupling (S_{21})	-15 to -30 dB	-10 to -15 dB	-10 to -15 dB	-5 to -15 dB
Grating Lobes	Occurs when $d \geq \lambda$	Limited array use	Limited array use	Strong when $d \geq \lambda$

In light of various studies by researchers such as Balanis, Hansen, Gazit, and Schantz, Vivaldi Antennas show better performance in terms of controlled grating lobes for a spacing of less than one wavelength, reduced mutual coupling, and impedance stability. Hence, Vivaldi Antennas are suitable for Ultra-Wideband Array Antennas, and this paper aims to perform a detailed analysis of Vivaldi Antennas.

II. ANTENNA DESIGN PARAMETERS

The designed antenna element, which is based on the concept of the tapered slot Vivaldi antenna, has been designed at a centre frequency of 6 GHz using the MATLAB antenna toolbox. The free-space wavelength corresponding to this operating frequency is 50 mm. The antenna element has been chosen for its ultra-wideband characteristics and stable radiation performance. The antenna element can also be used for array configurations. The geometrical and simulation parameters of the antenna element and the array configuration are listed in Table 2.

TABLE 2

Parameter	Value
Operating frequency (fc)	6 GHz
Free-space wavelength (λ)	50 mm
Antenna type	Tapered-slot Vivaldi antenna
Radiation type (single element)	End-fire
Radiation type (array)	Broadside
Array configuration	Linear array
Number of elements analysed	1, 4, 8, 14
Inter-element spacing (d)	0.8λ , λ , 1.2λ
Excitation type	Uniform excitation
Progressive phase shift	0°
Simulation tool	MATLAB Antenna Toolbox
Mesh size	$\lambda/8$

From Fig. 1, it is seen that the antenna elements are along the z-axis. For an isolated Vivaldi element, Fig. 2 reveals that the radiation is primarily end-fire-the maximum radiation occurs in the direction of the longitudinal axis of the element, i.e. along the z-direction. This can be interpreted from the fact that the primary lobe is oriented along the z-axis and thus the operation for the isolated element is end-fire.

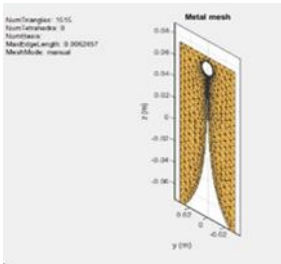


Fig. 1. Single-element Vivaldi

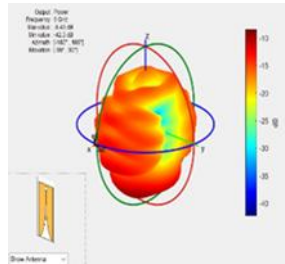


Fig. 2. Elemental radiation pattern

The radiation pattern for the Vivaldi array is shown in Fig. 4. From this, one finds that it possesses a broadside radiation characteristic, as the maximum radiation occurs perpendicular to the longitudinal axis. This shows that the combined effect of the pattern of the individual element and the array factor influences the overall radiation pattern for a Vivaldi antenna array. If the elements in the array are excited with equal amplitudes and zero progressive phase shift [2], [4], there will be constructive interference in the broadside direction where the array factor will be maximum. Consequently, even though each individual element inherently supports end-fire radiation, the total radiation pattern of the linear Vivaldi array exhibits a broadside maximum.

In arrays:

$$\text{Total Pattern} = \text{Element Pattern} \times \text{Array Factor}$$

To investigate the effect of array size, Vivaldi antenna arrays with different numbers of elements were analyzed while maintaining uniform inter-element spacing. As observed from Fig. 2 and Fig. 4, increasing the number of array elements significantly improves gain and directivity while narrowing the beamwidth [8]. Larger arrays concentrate radiated energy more effectively in the main lobe and suppress sidelobe levels, resulting in improved long-range performance.

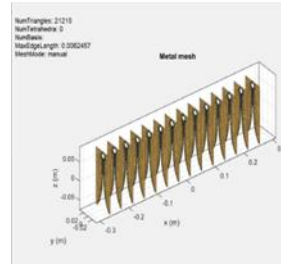


Fig. 3. 14-element array

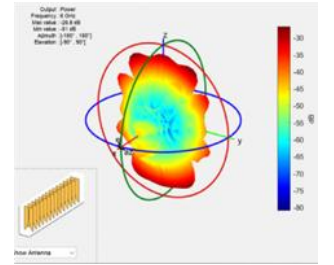


Fig. 4. 14-element array radiation pattern

III. IMPACT OF ANTENNA SPACING IN VIVALDI ARRAY

From Fig. 5, for $d = 0.8\lambda$, the radiation pattern remains stable with a clean main lobe. When spacing increases to $d = \lambda$, minor sidelobe variations appear. However, for $d = 1.2\lambda$, strong grating lobes dominate the radiation pattern, degrading directional performance. For a uniform linear array, the directions of maximum radiation are determined by the array factor condition

$$k d \sin \theta = 2\pi m, m=0, \pm 1, \pm 2, \dots$$

where $k = \frac{2\pi}{\lambda}$, d is the inter-element spacing, and θ is the angle of observation.

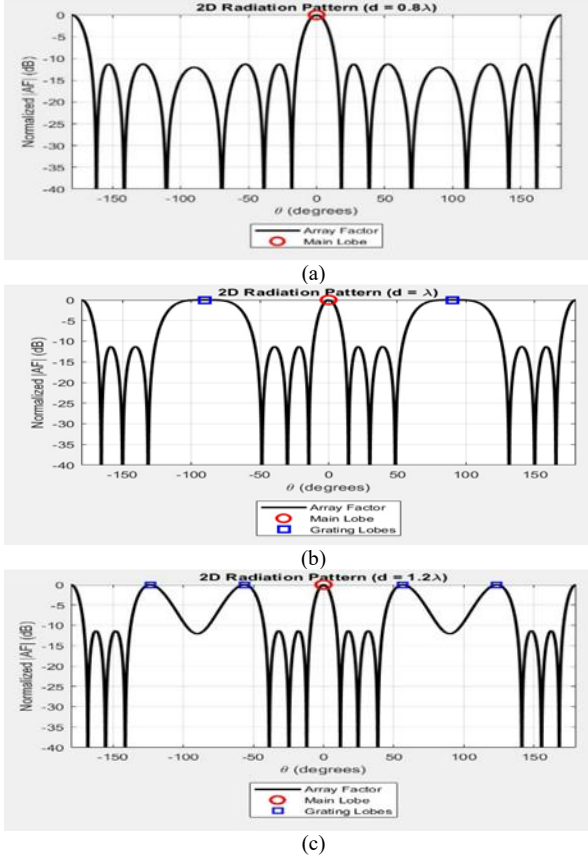


Fig. 5. Radiation patterns of a 4-element Vivaldi array for different inter-element spacings. (a) $d=0.8\lambda$ (b) $d=\lambda$ (c) $d=1.2\lambda$

When the element spacing is equal to one wavelength ($d = \lambda$), the above condition reduces to

$$\sin \theta = m$$

The solution $m = 0$ corresponds to the main lobe of a broadside array at $\theta = 0^\circ$. The solutions $m = \pm 1$ give rise to two symmetric grating lobes located at $\theta = \pm 90^\circ$, which lie at the edges of the visible region. Therefore, there are exactly two grating lobes for $d = \lambda$, due to the symmetry of the sine function. For element spacing $d < \lambda$, No real solutions exist for $m \neq 0$, and hence grating lobes are suppressed. For element spacing $d > \lambda$, multiple real solutions occur, resulting in additional grating lobes within the visible region which degrade array performance [2], [3], [4].

The spacing $d = \lambda$ is referred to as the critical spacing, beyond which grating lobes begin to appear.

A. Vswr characteristics

Fig. 6 shows that spacing $d = \lambda$ provides the most stable impedance matching with VSWR remains below 2 across most of the operating band [12]. Smaller spacing suffers from strong coupling, while larger spacing degrades matching due to grating lobe formation.

$$vswr = \frac{1 + |S_{ii}|}{1 - |S_{ii}|}$$

Where S_{ii} is the reflection coefficient

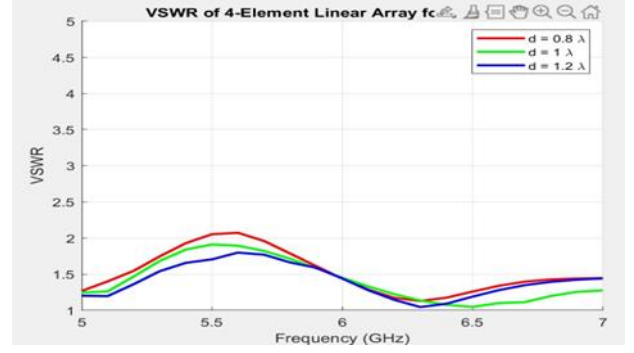


Fig. 6. VSWR characteristics of a 4-element Vivaldi array for different element spacings.

B. Input impedance characteristics

An antenna array has an input impedance, which can be described as $Z_{in} = R + jX$.

$$Z_{in} = Z_o \times \left(\frac{1 + S_{ii}}{1 - S_{ii}} \right)$$

Where,

Z_o = Characteristic impedance

For the best performance, we want the input resistance to match the system's characteristic impedance closely, while the reactance should be near zero.

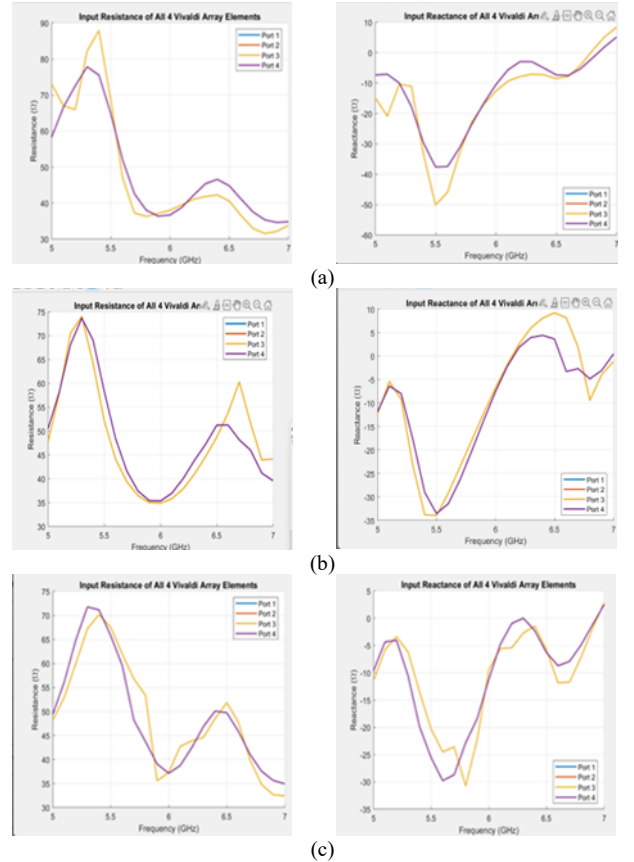


Fig. 7. Resistance and reactance plots of the 4-vivaldi antenna array for different element spacings (a) $d=0.8\lambda$, (b) $d=\lambda$, (c) $d=1.2\lambda$

Negative reactance indicates capacitive behavior (short antenna), while positive reactance indicates inductive behavior (long antenna). Zero reactance corresponds to resonance and maximum power transfer.

C. Grating lobes analysis

The input impedance remains close to the ideal $50 \Omega (Z_0)$ resistive condition with minimal reactance over the operating band. Symmetry in the array results in identical impedance behavior for element pairs (1,4) and (2,3) in a 4-element array.

Evaluation of Grating Lobe Performance and Charting Grating Lobe Performance for a Broadside Vivaldi Antenna Array at $\theta=0^\circ$ for a Broadside Vivaldi Antenna configuration.

Element spacing shall adhere to the following equation for a given Array Design ($\theta=0^\circ$)

$$d < \frac{\lambda}{1 + |\sin \theta|}$$

Where:

d = Element Spacing in meters

λ = Free-space Wavelength of the highest frequency in MHz

θ = Maximum Scan Angle in degrees

For Broadside Vivaldi Antenna Arrays ($\theta=0^\circ$) $d < \lambda$ for Grating Lobe Suppression [13].

Element spacing less than 1-Wavelength guarantees the absence of Grating Lobes in an antenna's Main Lobe region. Decreasing element spacing also enhances Grating lobe suppression; however, this equates to increased element-to-element coupling. Industry-Standard for Element Spacing of Ultra-wideband Vivaldi Arrays is $d \leq 0.8\lambda$ to achieve optimum balance between coupling & Bandwidth [10]. Spacing Near $d = \lambda$ typically identifies where Grating Lobes begin occurring in the antenna's performance graph above higher frequencies. For scanning applications, $d \leq \lambda/2$ is preferred to completely avoid grating lobes.

D. Mutual coupling analysis

Mutual coupling in the Vivaldi array is dominated by inter-element spacings. When d is 0.8λ , this leads to a strong near-field coupling effect between the neighbouring elements, causing degradation in the VSWR and a slight increase in the main lobe size, although grating lobes are suppressed due to close inter-element spacings [14].

In the array with $d = 1.0\lambda$, the coupling between the array elements is mitigated because of the array spacing, resulting in better impedance matching ($VSWR < 2$) along with a fixed main lobe. Coupling effects on the side lobe level are evident with minor fluctuations.

For $d = 1.2\lambda$, the effect of mutual coupling is insignificant, but the long separation introduces spatial aliasing, with high grating lobes [15]. These effects overcome the effect of the reduced coupling.

Mutual coupling approximately decreases exponentially with spacing [11]:

$$|S_{ij}| \propto e^{-\alpha d}$$

where:

d = inter-element spacing

α = attenuation constant (frequency dependent)

Mutual coupling can also be reduced by incorporating defected ground structures (DGS) or electromagnetic bandgap (EBG) structures between adjacent antenna elements to suppress surface-wave interactions. Additionally, the use of parasitic isolating elements or polarization diversity between neighbouring elements further improves isolation without significantly increasing array size.

IV. IMPACT OF NUMBER OF ELEMENTS AT FIXED INTER-ELEMENT SPACING

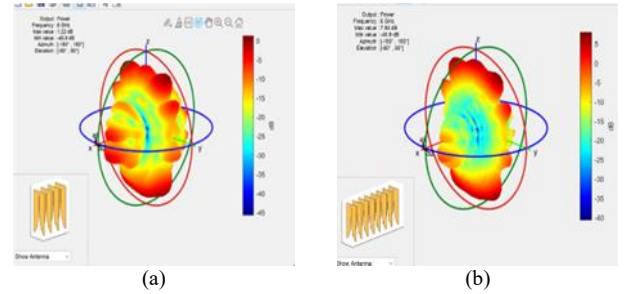


Fig. 8. Radiation patterns of Vivaldi arrays with different numbers of elements at fixed spacing ($d = 0.8\lambda$) (a) $N=4$ (b) $N=8$

Increasing the number of elements at constant spacing ($d=0.8\lambda$) enhances gain and directivity while maintaining grating-lobe-free radiation. However, mutual coupling accumulates, increasing capacitive loading and reducing array stability.

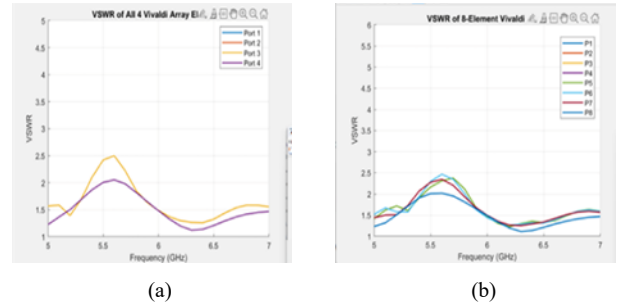


Fig. 9. VSWR of Vivaldi arrays with different numbers of elements (a) $N=4$ (b) $N=8$

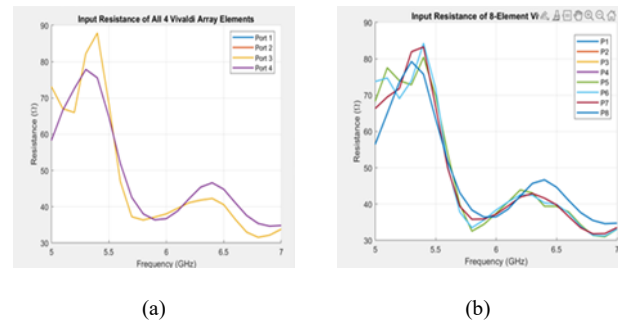


Fig. 10. Input Resistance of Vivaldi arrays with different numbers of elements (a) $N=4$ (b) $N=8$

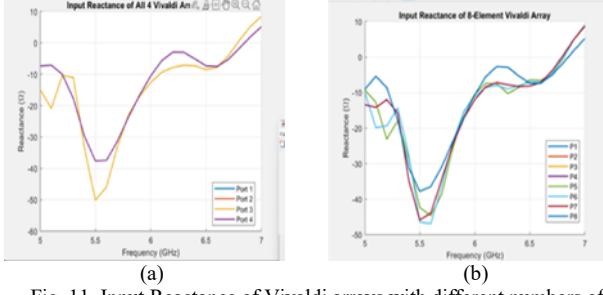


Fig. 11. Input Reactance of Vivaldi arrays with different numbers of elements at fixed spacing

As the array size increases, capacitive loading increases due to accumulated mutual coupling [16].

$$X_{active} = X_{self} + \Sigma X_{mutual}$$

Table 3 determines the effect of the number of elements in an array on various parameters.

TABLE 3

Parameter	4-Element Array	8-Element Array	Effect of Increasing Number of Elements
Directivity/ Gain	Medium gain ($\approx 12\text{--}15$ dBi)	Very high gain ($\approx 17\text{--}20$ dBi)	Gain increases with array aperture
Beamwidth	Wide beamwidth ($\approx 40^\circ\text{--}50^\circ$)	Narrow beamwidth ($\approx 20^\circ\text{--}30^\circ$)	Beamwidth decreases
3D Radiation Pattern	Moderately focused broadside pattern	Highly focused broadside pattern	Radiation becomes more directional
Grating Lobes	Suppressed (for $d = 0.8\lambda$)	Suppressed (for $d = 0.8\lambda$)	Proper spacing prevents grating lobes
Front-to-Back Ratio	Fair ($\approx 10\text{--}15$ dB)	Excellent ($\approx 18\text{--}25$ dB)	Forward radiation improves
Impedance Matching	Good (VSWR < 2)	Acceptable (VSWR $\approx 1.8\text{--}2$)	Slight degradation due to coupling
Stability	Moderate	Slightly lower	Stability slightly reduces
Mutual Coupling Effect	Lower (≈ -18 to -22 dB)	Higher (≈ -12 to -18 dB)	Coupling increases with elements
Array Complexity	Lower feeding complexity	Higher feeding complexity	Feeding complexity increases
Overall Performance	Best balance	Best gain and directivity	Performance improves with trade-offs

V. CONCLUSION

The UWB Vivaldi antenna arrays simulated and operating at 6 GHz were analyzed in the aspects of impedance matching, VSWR, radiation patterns, mutual coupling and grating lobes. Compared with spiral, sinuous and coupled antennas, the Vivaldi antenna has better performance of wideband impedance matching, stable broadside radiation and array fabrication. Enhancing the number of array elements at 6 GHz not only contributes to gain and directivity enhancement, but also to a decrease in beamwidth and sidelobe levels while increasing mutual coupling. Consequently, a well-designed UWB Vivaldi array at 6 GHz demonstrates the advantages of high gain and stable impedance matching as well as grating-lobe-free radiation, which is applicable to wideband communication and UWB localization systems

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